Brandeis
University Division of Graduate
Professional Studies Rabb
School of Continuing Studies

Course syllabus

I. Course Information

1. **Course Name: Biomedical Statistics with R**
2. **Course Number: RBIF 111**
3. **Course weeks: 8 course weeks.**
4. **Instructor's Name and Contact Information**

Brendan Gongol PhD
Email:brendangongol@brandeis.edu
Office hours:
Tuesday from 12:00 PM - 1:00 PM EDT (9:00-10:00 PDT).
Thursday during the lunch hour from 12:00 PM - 1:00 PM EDT (9:00-10:00 PDT) and at the end of the day 6:00 PM-7:00 PM EDT (3:00 PM-4:00 PM PDT).
Office hours are held at the following site: https://brandeis.zoom.us/j/99913604201

5. **Document Overview**

   This syllabus contains all relevant information about the course: its objectives and outcomes, the grading criteria, the texts and other materials of instruction, and of weekly topics, outcomes, assignments, and due dates.

   Consider this your roadmap for the course. Please read through the syllabus carefully and feel free to share any questions that you may have. Please print a copy of this syllabus for reference.

6. **Course Description**

This course is an advanced mathematics and applied statistics course that will introduce students to data analysis methods and statistical testing. It provides a foundation for Biological Data Mining and Modeling (RBIF 112) and Design and Analysis of Microarray Experiments (RBIF114). This course will use R (a statistical programming language) to introduce students to descriptive and inferential statistics, basics of programming, common data structures and analysis techniques. It will cover methods important to data analysis such as t- tests, chi-squared analysis, Mann-Whitney tests, correlation and regression, ANOVA, PCA, tests of significance, Fisher's exact test, etc.

7. **Materials of Instruction**

   a. **Recommended Texts**

   The following two texts are not required for the course; both are available for free online and may be useful as references during the course and over the course of your career.

   Book 1: An Introduction to Statistical Learning with Applications in R

   Author: Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani

Publisher: Springer, ISBN: 978-1-0716-1417-4 (Hardcover) / 978-1-0716-1420-4 (Softcover) / 978-1-0716-1418-1 (eBook), Edition: 2nd Edition (2021), (free PDF available at https://www.statlearning.com/)

Book 2: R for Data Science

Author: Hadley Wickham, Mine Çetinkaya-Rundel, Garrett Grolemund

Publisher: O'Reilly Media, ISBN: 978-1-492-09740-2 (Print), Edition: 2nd Edition (2023), (free online at https://r4ds.hadley.nz/)

Book 3: Hierarchical Linear Models: Applications and Data Analysis Methods. 2nd ed.

Author: Raudenbush, Stephen W., and Anthony S. Bryk.,

Publisher: SAGE Publications, 2002.

(free online at: https://eli.johogo.com/Class/CCU/SEM/_Hierarchical%20Linear%20Models%20Applications%20and%20Data%20Analysis%20Methods_Raudenbush.pdf)

b. Required Software and Tools

- R software, free and available from CRAN: http://cran.r-project.org/
- Highly recommended: R Studio (free edition) – a very good IDE for R, a de facto industry standard

c. Online Course Content

This is a Distance Learning (DL) course, which will be hosted at Brandeis' moodle site, available at

https://bonline.brandeis.edu/my/courses.php

The site contains the course syllabus, weekly topic notes, assignments, and discussion forums through which we will communicate during this course.

8. Development Emphasis Project (Optional)

Students have the opportunity to contribute to an in-development open source R package as a development emphasis project, a voluntary, extracurricular activity. Each contribution must implement a statistical method from the course as a checked function that runs correctly on bulk, large-scale data. Contributions are not required and are not graded, but offer additional mentorship opportunities. This project pairs naturally with the statistical emphasis in this course and the data wrangling and reproducible-workflow skills covered in R for Data Science above. Full details on contribution tiers, the proposal and review process, and how to get started are provided in the Week 1 Notes, Part 5 (Development Emphasis Project).

9. Overall Course Objectives

The course is intended to provide students with working knowledge of:

R Programming: basic and advanced analysis and graphing techniques Approaches for data analysis with linear algebra
Statistical modeling
Formal inference and estimation
Contingency tables and Fisher's exact test Confidence intervals and hypothesis tests Correlation and regression
Multiple Linear Regression
Linear models

Analysis of variance
Non-parametric methods

Survival analysis (Cox proportional hazards, accelerated failure time, and Gompertz models)
Advanced modeling for causal inference and longitudinal data (propensity score matching, instrumental variables, mediation analysis, mixed-effects models, generalized estimating equations, inverse probability weighting, multiple imputation, ROC analysis, and random-effects meta-analysis)
Penalized regression for high-dimensional prediction (Ridge, Lasso, and elastic net) and partial correlation analysis
Reproducible research workflows (R Markdown/Quarto, DuckDB) and GitHub-based version control for professional portfolio development
Opportunity to contribute to open source R package development as a development emphasis project, a voluntary extracurricular activity that involves implementing a statistical method as a checked, bulk-data function, building verifiable professional credentials including vignette contributor credit, an "R Package Developer" CV line, and potential co-authorship on peer-reviewed publications

10. Overall Course Outcomes

At the end of the course, students will be able to:

1. Write programs in R for data analysis
2. Discern appropriate data models to answer specific questions
3. Analyze data with a number of statistical models
4. Conduct explorative analysis of large data sets
5. Choose appropriate methods of analysis for a given problem
6. Perform tests to validate models against data
7. Evaluate various statistical methods on the basis of their strengths and weaknesses
8. Be resourceful in identifying and using available tools, including AI, to deepen understanding and solve problems independently
9. Take advanced courses on the analysis of biological data (RBIF112 and RBIF114)

11. Course Grading Criteria

Percent	Component
60%	Homeworks, averaged (a total of 8 homeworks will be assigned)
40%	Discussions/online participation: includes original responses and replies in the discussion forum

1. Students will be assigned homework on a weekly basis: 8 homework assignments over the term.
2. Final exam: last week of the classes.
3. Mid-term examination: fifth week of the classes.
4. Class participation is graded based on student's replies to discussion questions and substantive replies to other posts on the class forum. Students are expected to read weakly course content and are required to
    - Post at least one original substantive response to each of the two discussion questions offered each week; the first response is due by Saturday midnight EST and the second response is due by Monday midnight EST
    - Post at least two substantive replies to the posts of others in the discussion question forums by Tuesday midnight EST
    - Make their required postings on at least three different days.

Participation Evaluation: Each week, 100 raw points may be earned toward the participation

component of the grade.

Original Responses: Maximum raw points earned for original responses each week: 30 points each, 60 points total. Evaluation Criteria: Includes your own insights into the topics, sharing your professional experiences as appropriate and your own conclusions; Includes references to weekly required readings and/or other external sources, cited appropriately; Answers the question posed completely; Consists of at least 200-300 words; Well written, with no spelling or grammatical errors. Original responses one day late: no credit

Substantive Replies: Maximum raw points earned for substantive replies: 15 points each, 30 points total. Evaluation Criteria: Substantive (beyond an "I agree" or complimentary post) with follow-on points from your related experiences and/or from the readings and follow-up questions of others to extend the conversation (encouraged); consists of at least 200 words. Replies one or more days late: no credit.

Posting on Three Days of the Course Week

- Post on three days of the course week: 10 points (full credit)
- Post on two days of the course week: 6 points
- Post on one day of the course week: 2 points

12. Rubrics

Grading Rubric for Assignments

Your assignment grades are based on the following criteria:

Criteria	A,A- (90-100)	B+,B,B- (80-89)	C+,C,C- (70-79)	D+,D,D-,F (69 or below)
Correctness (30%)	The answers are correct according to the answer key	There are a few minor differences between the answers	There are major differences between the answers and the answer key	Answers are missing or not attempted
Code Completeness (30%; includes GitHub upload where an assignment requires it)	The code is well-written and executes correctly. Where the assignment requires it, the code and rendered HTML are both present in the linked GitHub repository.	The code executes but contains minor mistakes, or the GitHub repository (where required) is missing one of the two required files (code or rendered HTML)	There was an attempt to write the code, but major mistakes were made which prevents it from executing, or the required GitHub repository link is missing or inaccessible.	Major portions of the code are missing and/or the answer was not attempted.
Critical thinking (30%)	Results are clearly interpreted and explained; if AI was used, the submission shows clear evidence of understanding (e.g., the reflection note demonstrates	Results are interpreted but lack key concepts.	The explanation of the results is fundamentally incorrect.	An explanation of the results was not attempted.

	genuine learning, not just a restated answer)			
Format (10%)	The assignment is submitted as a knitted and raw markdown file.	The materials are submitted, but the markdown file is not knitted.	A markdown file was not used and/or the answer is provided in several different files.	Files are missing and/or the assignment was not submitted.
Meets deadline (5% reduction of overall grade for each missed day)	The assignment is submitted on time or 1 day late	The assignment is submitted 2 days late.	The assignment is submitted 3 days late	The assignment is submitted 4 days late (no credit will be provided for assignments submitted more than 7 days late).
AI Reflection (if AI was used; see Section VI, AI Policy)(5%)	A note is included describing what was learned, what was verified or corrected, or how the AI's output was evaluated before use. The note shows genuine engagement, not just a restatement of what AI said.	A note is included but only restates what AI said without describing how it was evaluated, verified, or what the student now understands differently.	AI was used but no reflection note was included.	There is no note, and no reflection about what was learned from AI.

Note: the AI Reflection row is not a compliance checkbox. It is an opportunity to demonstrate that you used AI as a learning tool rather than a shortcut. A strong reflection describes what you verified, what you corrected, or what you now understand differently as a result of engaging critically with AI output. See Section VI (AI Policy) for guidance on how to use AI well in this course.

II. Weekly Information

Week 1	
Outcomes	Be able to load, plot and save data in R, use common data structures, functions and operators, access R documentation, and download and manage open-source biomedical data (including GEO and SummarizedExperiment containers). Be able to generate DuckDB databases with parquet files for back end data storage, and produce reproducible reports using R Markdown or Quarto. Be able to use GitHub for version control, including branching, pull requests, and code review, as a course-wide practice for building a professional portfolio. Understand the voluntary development emphasis project offered through this course,

	including the requirement to implement a statistical method as a checked, bulk-data function, the contribution tiers, the proposal and review process, and the professional credentials available to contributors.
Assignments	Homework wk 1

Week 2	
Outcomes	Be able to visualize data and inspect data plots to reveal patterns and irregularities in data; understand concept of null hypothesis, plan data analysis, use simple resampling simulation and t-test. Be able to assess normality using QQ plots and the Shapiro-Wilk test, and apply non-parametric alternatives (e.g., the Wilcoxon/Mann-Whitney test) when normality assumptions are not met.
Assignments	Homework wk 2

Week 3	
Outcomes	Be able to fit simple linear models in R and test whether model assumptions are defensible; use probability distributions and random numbers; understand statistical independence and maximum likelihood estimation. Be able to apply non-parametric counterparts to linear methods (e.g., Spearman rank correlation, Wilcoxon rank-sum test, Wilcoxon signed-rank test for paired data, and Kruskal-Wallis test) when linear model assumptions are violated, and understand when to prefer non-parametric over parametric tests. Be able to apply non-parametric regression smoothers (LOESS/LOWESS) and spline regression to visualize and model nonlinear relationships, and fit generalized additive models (GAMs) using the mgcv package to estimate flexible smooth functions for each predictor while retaining additive model interpretability and supporting formal significance testing.
Assignments	Homework wk 3

Week 4	
Outcomes	Be able to use contingency tables and analysis of variance, calculate confidence intervals and interpret results of hypotheses' tests. Be able to test for association in contingency tables using both the chi-square test and Fisher's exact test, and understand when each is appropriate. Be able to apply time series methods to sequentially measured biomedical data, including decomposition into trend and seasonal components, ARIMA models for modeling autocorrelated data and forecasting, and interrupted time series analysis for detecting the effect of a known intervention.
Assignments	Homework wk 4

Week 5	
Outcomes	Be able to fit and interpret one-way and two-way analysis of variance (ANOVA) models, and apply contingency table analysis (building on Week 4) to more complex designs. Be able to fit generalized linear models, including logistic regression, for non-normally distributed outcomes. Be able to conduct survival analysis, including Cox proportional hazards models, accelerated failure time (AFT) models, and Gompertz models, and compare the assumptions and tradeoffs of semi-parametric versus parametric approaches to time-to-event data. Be able to design and interpret equivalence and non-inferiority tests, and calculate and correctly interpret absolute risk reduction (ARR), relative risk reduction (RRR), number needed to treat (NNT), and

	number needed to harm (NNH). Be able to formally test for treatment-by-subgroup interactions and explain why within-subgroup significance comparisons are not sufficient evidence of differential treatment effects.
Assignments	Homework wk 5

Week 6	
Outcomes	Be able to use permutation testing and bootstrap resampling to estimate significance and confidence intervals without relying on parametric assumptions. Be able to apply multiple testing correction methods, including the Bonferroni correction, Benjamini-Hochberg false discovery rate (FDR), and permutation-based FDR estimation, when testing many hypotheses simultaneously. Be able to design and interpret planned interim analyses in clinical trials using group sequential methods, alpha-spending functions (O'Brien-Fleming, Pocock), and futility boundaries, and explain how repeated testing of accumulating trial data inflates the Type I error rate.
Assignments	Homework wk 6

Week 7	
Outcomes	Be able to fit linear models with multiple predictors, identify common problems with model fitting, assess and compare multiple models. Be able to assess predictive accuracy using cross-validation (including train/test split and k-fold cross-validation) and bootstrapping, and interpret the design matrix underlying a fitted model. Be able to fit penalized regression models (Ridge, Lasso, elastic net) for high-dimensional prediction problems where the number of predictors exceeds the number of samples, select hyperparameters via cross-validation, and interpret the resulting sparse coefficient sets. Be able to compute and interpret partial correlations between variables after adjusting for one or more confounders, and explain the relationship between partial correlation and multiple regression coefficients.
Assignments	Homework wk 7

Week 8	
Outcomes	Be able to apply advanced statistical modeling methods for biomedical data, including propensity score matching, instrumental variables, and mediation analysis for causal inference; mixed-effects models and generalized estimating equations (GEE) for repeated-measures and longitudinal clinical trial data; inverse probability weighting (IPW) and multiple imputation for missing data; ROC analysis for evaluating classifier performance; and random-effects meta-analysis (REML) for pooling effect size estimates across independent studies. Be able to apply generalized additive models (GAMs) for covariate adjustment, mixing smooth terms for nonlinear covariates and linear terms for binary or linear covariates in the same model, and use edf and AIC to decide whether a linear or smooth adjustment is appropriate for each covariate. Be able to distinguish intention-to-treat (ITT) from per-protocol (PP) analyses, explain why ITT is the regulatory standard for confirmatory trials, and apply multiple imputation correctly in the ITT context to handle informative dropout.

Assignments	Homework wk 8

III. Course Policies and Procedures

1. Late Policies

All homework (mid-term exam is homework!) is due at midnight on the last day of the course week (Tuesday).

Homework submitted after that time will be deducted 10% points for each day late for three days. Assignments not submitted after three days will not receive credit.

All discussion posts are due by the closing of each week. No late credit is applied to discussion posts submitted after the deadline.

2. Grading Standards

- All work will be expected to offer both computer programs executed as well as results, as required in the homework assignment document.
- No outside R programs may be used. Plagiarism is not allowed and University policies on plagiarism are strictly in force. If you have any questions on the appropriateness of a reference or a computer program you would like to use, please contact the instructors before handing your homework in or proceeding using others' work. It is better to be safe than sorry.

How Points and Percentages Equate to Grades:

100-94	A		76-73	C
93-90	A-		72-70	C-
89-87	B+		69-67	D+
86-83	B		66-63	D
82-80	B-		62-60	D-
79-77	C+		59 or <	F

Feedback

Feedback will be provided on assignments and the exam within 10 days of the due date. Within 7 days of the close of each week, feedback will be provided on weekly participation via the associated Weekly Participation Feedback assignments; you do not have to submit anything to these assignments. I will be recording your weekly participation grades and providing narratives describing your discussion posts (participation) for that week.

3. Workload Expectations

- Students are responsible to explore each week's materials and submit required work by their due dates. On average, a student can expect to spend approximately 2-4 hours per week reading and approximately 6-9 hours per week completing assignments and posting to discussions. The calendar of assignments and due dates is located at the end of this syllabus, and all assignments are due by the close of the associated week (Tuesday evenings).

4. Access to the Course Site

Members of the University's technical staff have access to all course sites to aid in course setup and technical troubleshooting. Program Chairs and a small number of Graduate Professional Studies (GPS) staff have access to all GPS courses for oversight purposes. Students enrolled in GPS courses can expect that individuals other than their fellow classmates and the course instructor(s) may visit their course for various purposes. Their intentions are to aid in technical troubleshooting and to ensure that quality course delivery standards are met. Strict confidentiality of student information is maintained.

IV. Github

Select course assignments include a voluntary portfolio development exercise (see, for example, the Week 1 assignment). These exercises are not mandatory and carry no point value. They are an opportunity to build out a professional portfolio for students who choose to complete them. Where a question specifies a GitHub submission, such as this portfolio development exercise, you are required to push your code (the .Rmd file) and your rendered HTML output to a GitHub repository and include the repository link in your submission. The other, required questions in that same assignment do not require a GitHub submission. You will also submit these documents to the grading portal. The intent of this requirement is for your repository to be something you can show to a future employer, so make sure both files are present and the HTML renders your full analysis and output. Grading for the GitHub component of a given assignment is described in that assignment's instructions and rubric, and is incorporated into the Code Completeness criterion (see Section 12, Rubrics).

Beyond any specific assignment requirement, students are encouraged to utilize the information in this course to develop a github repository (https://github.com/) that showcases their work more broadly. Github offers the ability to post code that has been developed in both private and public settings. It is recommended that when posting to a public github repository that the code is polished, functionalized (There is one function per task such that lengthy code is not displayed), function files are sourced as opposed to loading functions at the beginning of a document, and there are sufficient explanations about the meaning and interpretation of the data, code, and results. In addition, posting answers to course problems prior to receiving assignment feedback is a source of academic misconduct and any plagiarism that is identified through this mechanism will be subjected to university Academic honesty and student integrity guidelines.

V. University and Division of Graduate Professional Studies Standards

Please review the policies and procedures of Graduate Professional Studies, found at http://www.brandeis.edu/gps/students/studentresources/policiesprocedures/index.html. Among them, we would like to highlight the following.

Learning Disabilities

If you are a student with a documented disability on record at Brandeis University and wish to have a reasonable accommodation made for you in this course, please contact me immediately.

Academic Honesty and Student Integrity

Academic honesty and student integrity are of fundamental importance at Brandeis University and we want students to understand this clearly at the start of the term. As stated in the Brandeis *Rights and Responsibilities* handbook, “Every member of the University Community is expected to maintain the highest standards of academic honesty. A student shall not receive credit for work that is not the product of the student’s own effort. A student's name on any written exercise constitutes a statement that the work is the result of the student's own thought and study, stated in the students' own words, and produced without the assistance of others, except in quotes,

footnotes or references with appropriate acknowledgement of the source." In particular, students must be aware that material (including ideas, phrases, sentences, etc.) taken from the Internet and other sources **MUST** be appropriately cited if quoted, and footnoted in any written work turned in for this, or any, Brandeis class. Also, students will not be allowed to collaborate on work except by the specific permission of the instructor. Failure to cite resources properly may result in a referral being made to the Office of Student Development and Judicial Education. The outcome of this action may involve academic and disciplinary sanctions, which could include (but are not limited to) such penalties as receiving no credit for the assignment in question, receiving no credit for the related course, or suspension or dismissal from the University.

Further information regarding academic integrity may be found in the following publications: "In Pursuit of Excellence - A Guide to Academic Integrity for the Brandeis Community", "*(Students') Rights and Responsibilities Handbook*" AND "Continuing Studies Student Handbook". You should read these publications, which all can be accessed from the Continuing Studies Web site. A student that is in doubt about standards of academic honesty (regarding plagiarism, multiple submissions of written work, unacknowledged or unauthorized collaborative effort, false citation or false data) should consult either the course instructor or other staff of the Rabb School for Continuing Studies.

University Caveat: The above schedule, content, and procedures in this course are subject to change in the event of extenuating circumstances.

VI. AI Policy

AI tools are a welcome and encouraged part of how you learn in this course. Used well, they are a resource much like office hours, a textbook, or a study group: a way to ask questions, get a second perspective, work through confusion, and build a better understanding of the material. Part of what this course aims to teach is resourcefulness: knowing how to find and use the tools available to you, AI included, to deepen your own understanding rather than to shortcut it. This course does not prohibit or restrict any particular use of AI, and there is no list of approved or forbidden tools. Instead, the expectation is straightforward: if you use AI as part of your work, take a moment to note what you learned from it. That note is for your own benefit as much as for the course's. The act of articulating what you took away from an AI exchange is itself part of the learning process.

The goal isn't to catch misuse. It's for you to leave this course with real statistical and bioinformatics competence. A student who uses AI without understanding what it produced will not be able to explain their results, defend their methodological choices, or extend the approach to a new problem, and the existing standards in this course (correctness, code completeness, and critical thinking, as described in the assignment rubric) are what surface that gap. Resourcefulness, including knowing how and when to use AI effectively, is itself a skill this course wants you to develop.

Getting the most out of AI: AI is most valuable in this course when it is used to deepen your understanding. Here are concrete patterns that work well. Ask AI to explain a concept a different way when the notes are not clicking, then close the conversation and work through an example yourself to test whether it landed. Ask AI to explain what a piece of your own code does line by line, and correct it where it gets something wrong. Ask AI to identify potential problems with a statistical approach you are considering, then evaluate whether those problems apply to your specific data. Use AI to generate a candidate idea for an open-ended question, then critically evaluate it, identify its limitations, and implement a corrected version yourself. In every case the goal is the same: you should come away from the exchange understanding something you did not before, and you should be able to explain in your own words why the result is correct or where it went wrong. If you find yourself pasting in output you cannot explain, that is a signal to stop and work through it rather than move on. The reflection note in your assignment is the opportunity to capture that moment: not just what AI said, but what you verified, what you corrected, and what you now understand that you did not before.

A note on AI and correctness: AI tools can be wrong, can miss the nuance of a statistical problem, or can produce something that runs without actually answering the question being asked. This is a reason to stay engaged with what AI gives you. Assignments continue to be graded on the same standards as always, correctness, code completeness, and critical thinking, per the assignment rubric, regardless of whether AI was involved at any stage. The most reliable way to do well is the same whether or not you use AI. Understand the problem, understand your solution, and be able to explain both.

A note on academic honesty: Using AI thoughtfully and reflecting honestly on what you learned from it is, itself, the standard this course asks of you. The same trust extended to how you use any other resource, from textbooks to classmates to office hours. The university's general academic honesty standards (Section V of this syllabus) continue to apply to all coursework, AI-assisted or not, but this section does not impose any AI-specific restriction beyond what is described above.